

COM-480 - Milestone II - Blue Zones

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<https://com-480-data-visualization.github.io/BlueZones/>

1. Project Goal

As a brief recap, Blue Zones are five geographic regions (Okinawa, Ikaria, Sardinia, Nicoya Peninsula, and Loma Linda) where populations tend to live significantly longer than the global average. This exceptional longevity has been associated with nine lifestyle principles, commonly known as the “[Power 9](#)”. **Our project’s goal is to identify which countries have lifestyle patterns similar to those observed in Blue Zones and to evaluate whether there is a correlation between these patterns and life expectancy.**

Note: Since not all principles can be easily translated into measurable variables (e.g. having a strong sense of purpose), our analysis focuses on five key factors: natural movement, stress reduction (measured as happiness), “80% Rule”, plant-based diet, and wine consumption.

2. Main Components

- **Introductory Video:** The website will begin with a short, immersive video of the Blue Zones' environment that serves as a storytelling device, capturing user attention and providing a smooth transition into the core analytical visualizations.
- **Presentation of Blue Zones’ pillars:** Users can explore the nine Blue Zone pillars via an interactive force-directed chart. Selecting individual pillar bubbles reveals detailed explanatory cards. (This visual component will replace the current text block in the next milestone).

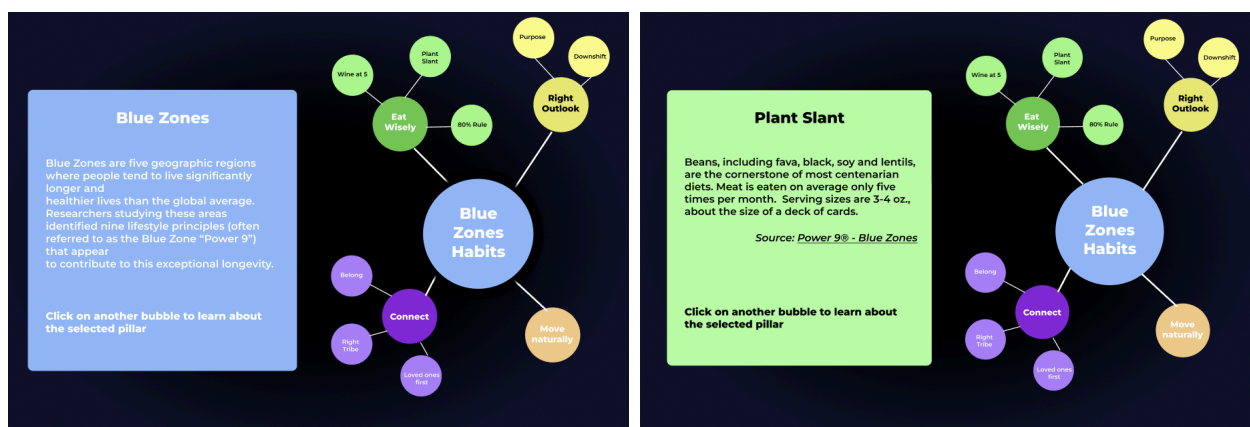


Figure 1 & 2: Force-directed chart with an explanatory card on Blue Zones (right), alongside the same visualization displayed when “Plant Slant” is selected (left)

- **Blue Zone Cards:** The Blue Zones will be presented as an automated carousel of interactive cards, each initially displaying an image of the corresponding region. As the carousel cycles through the cards, the currently visible region is simultaneously highlighted on the globe in real time. Users can also interact with any card by flipping it to reveal key information on the reverse side. This design creates a seamless connection between the carousel and the map, reinforcing the geographic context of each region.

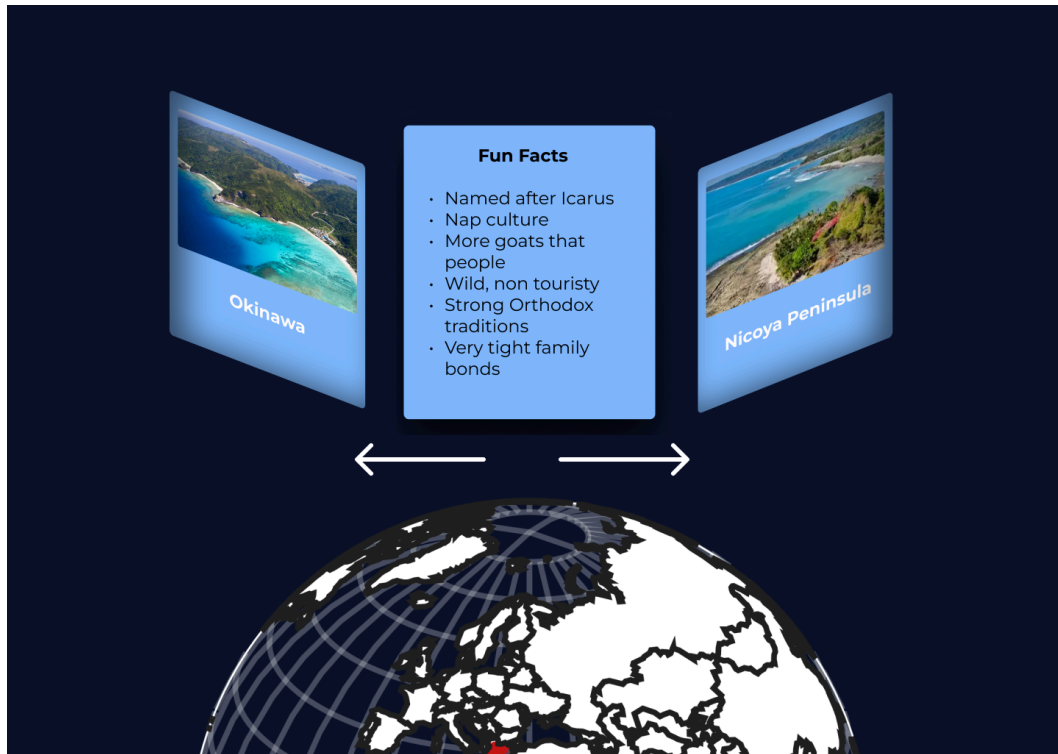


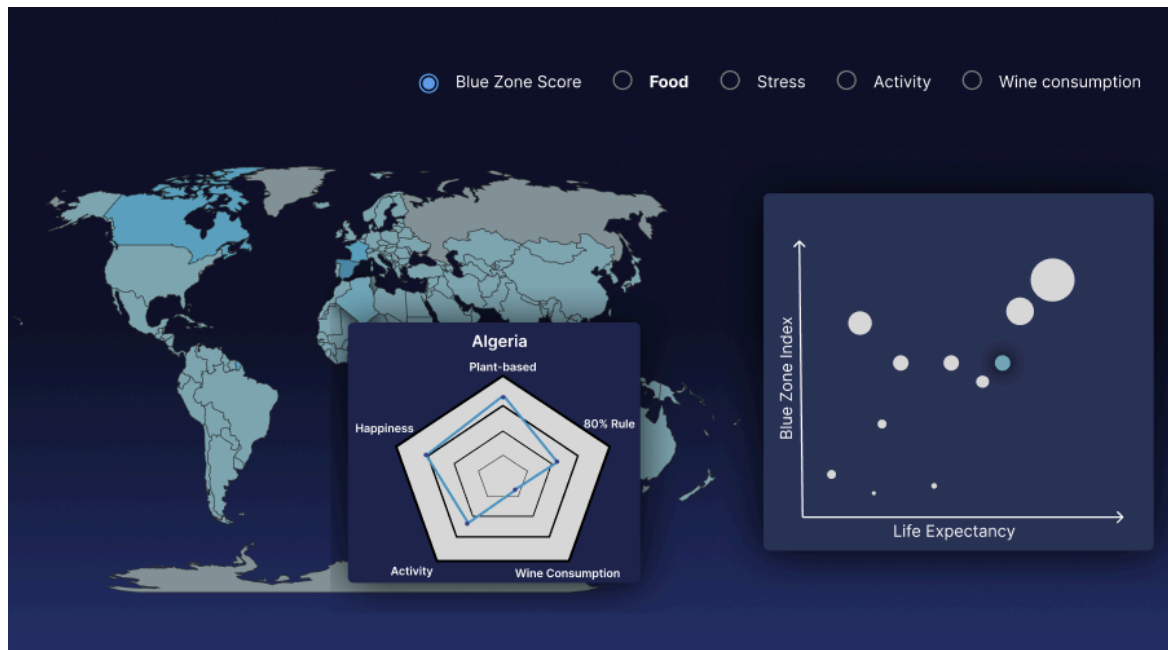
Figure 3: Blue Zone presentation cards and a globe showing these locations

- **Our approach:** A concise narrative section outlining the project's core objectives and our motivation for exploring the Blue Zones phenomenon.
- **World Map (MVP):** This will be the main visualization of our project, displaying a global map where each country will be represented using a color gradient from low to high alignment with Blue Zone lifestyle patterns. Using a set of radio buttons, users will be able to explore individual pillars or switch to a combined “Blue Zone Score,” which will summarize all factors into a single index, with the colors on the map updating accordingly.

When a country is selected, a radar (spider) chart will appear, showing its performance across the five pillars. This will allow users to quickly understand whether a country performs consistently well or stands out only in specific areas. For now in the demo website available on github only random data were used but this will also change in the next milestone.

In addition, a complementary scatter plot alongside the map will display the relationship between the “Blue Zone Score” and life expectancy. Each country will be

represented as a point, with life expectancy on the x-axis and the Blue Zone Score on the y-axis. When a user hovers over or selects a country on the map, the corresponding point will be highlighted in the scatter plot, showing its exact life



expectancy and creating a strong visual link between the two views.

Figure 4: Interactive world map with radio button controls and additional features

- Quiz:** The interface will feature a panel displaying a question with multiple-choice answers presented as radio buttons. Once the user selects an option, an information panel will appear, providing additional context or interesting facts. At the end, a score summary will be displayed, assigning a mastery level based on performance (e.g., "You are a Blue Zones Master").

Which of these is NOT one of the five Blue Zones?

☐ Okinawa, Japan

☐ Sardinia, Italy

☒ Reykjavik, Iceland

☐ Ikaria, Greece

CORRECT

The five Blue Zones are Okinawa (Japan), Sardinia (Italy), Ikaria (Greece), Nicoya Peninsula (Costa Rica), and Loma Linda (California).

🌟 Fascinating Fact 🌟

While Iceland has impressive longevity statistics, it's not classified as a Blue Zone. Reykjavik's residents do enjoy long lives, but the specific cultural and dietary patterns differ from traditional Blue Zone characteristics.

NEXT

Figure 5: Blue Zones Quizz

3. Required tools and lectures

The dataset will be constructed using HTML, CSS, and JavaScript, allowing for dynamic access and interaction on the client side. For the data visualizations, we will use D3.js to create interactive, data-driven components such as the world map, the globe and the scatter plot. Additionally, we will use Python with Pandas to structure the data according to our needs and compute derived variables such as the Blue Zone Score.

We will draw inspiration from all the lectures, with particular emphasis on:

- **Lectures on D3.js** for learning how to build interactive, data-driven visualizations and manage dynamic updates within the DOM.
- **Lecture on maps**, as our core visualization is a world map and the lecture provides the necessary foundations for working with geographic data formats such as GeoJSON.
- **Lecture on interactions**, as our project involves linking multiple visualizations.

4. Additional ideas

- A “What If” simulator that will allow users to explore how changes in a country’s lifestyle affect its Blue Zone Score. By adjusting the five pillars through simple interactive inputs, users will see the country’s score update in real time, along with dynamic changes across all the visualizations such as the radar chart, enabling them to understand which factors have the strongest impact to the overall score.
- Users will be invited to reflect on their own lifestyle by entering their habits, which will be translated into a personalized “Blue Zone Score”. This profile will then be compared against countries worldwide to uncover where their way of living most closely aligns, revealing unexpected similarities across cultures and regions and making the experience feel truly personalized.